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Intrusion detection algorithm based on multi-scale feature fusion

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ABSTRACT

Network intrusion detection plays a crucial role in ensuring cybersecurity by promptly mitigating network attacks. However, existing deep learning methods have limited capabilities in capture network attack features and address class imbalances, resulting in low classification accuracy. This paper proposes a deep-learning intrusion detection model named FLSPPMRXt, which is built upon ResNeXt50. It enhances feature capture by improving the backbone convolution and introducing a multi-scale feature fusion module, including the Soft Pool layer. Meanwhile, focal loss is employed as the loss function to effectively mitigate the impact of class imbalance on classification accuracy. Furthermore, this method proposes a data visualization processing algorithm to provide an image representation that is more consistent with the feature nearest neighbor distribution. Experimental results show that the FLSPPMRXt model achieves 93.3 % and 95.2 % in overall classification accuracy and F1 score on UNSW_NB15 dataset, respectively. Compared with existing algorithms, such as the 2DCNN and RNN models, the method demonstrates superior network intrusion detection performance.

1. Introduction

With the rapid development of emerging technologies such as cloud computing, fog computing, edge computing, and the Internet of Things, the Internet has become indispensable in people's daily lives. However, this convenience also comes with substantial cybersecurity implications. The popularity of the Internet has made our lives more convenient, but it has also provided opportunities for cyber intruders. It is crucial to create an efficient cyber intrusion detection system in order to reduce the threat that cyber intrusions pose.

Intrusion detection systems (IDS) (Hinton et al., 2006) can be categorized into host-based and network-based systems, depending on the origin of the detection data. Among these, the Network Intrusion Detection System (NIDS) is an integral component of the network security protection system, which plays a pivotal role in detecting network attacks and initiating corresponding defensive measures. Its primary function lies in detecting and identifying illicit and malicious traffic, thus prompting security administrators to prevent cyberattacks and mitigate financial losses.

Intrusion detection (Anderson, 1980) technology has been proposed for over 40 years, and various intrusion detection technologies are emerging. However, the continuous advancements in network equipment and technology, coupled with the rapid growth of network data, pose new challenges for network intrusion detection. While traditional

machine learning methods have achieved some success in detecting network intrusions, the majority of them are shallow learning algorithms that heavily rely on feature engineering. As cyberattacks continue to evolve and mutate and the ability of bad actors to bypass the rule base increases, securing cyber information and communications becomes increasingly challenging. Hence, there is an urgent need to enhance network security protection capabilities to address the pressing issues in network security.

As the hottest technology nowadays, deep learning has excellent potential for network intrusion detection. The nonlinear activation layer can effectively extract the relevant features from the original data. A system that can self-learn and detect unknown attacks can be established by using the training set. After model training, the system demonstrates a significant effect in identifying unknown attacks and can be used to detect novel, unknown, and abnormal data.

When building deep learning networks with network intrusion detection algorithms, on the other hand, the traditional convolutional layer will have issues like gradient disappearance or gradient explosion when the network is too deep, which will lower the accuracy of the classification. To solve this problem, He et al. (He et al., 2016) proposed a residual network (ResNet) to solve the problem of performance degradation as the network layer deepens. This is why Wu et al. (Wu et al., 2020) suggested a deep neural network made of residual blocks. This network had a low rate of false alarms in intrusion detection and

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fixed the issue of deep networks not working well. The residual learning model can learn deeper features, protect the integrity of information, and effectively solve the problem of network degradation, which has certain advantages in the field of intrusion detection. Also, ResNeXt (Xie et al., 2017) is an improved model based on the residual model, ResNet. It can handle more data and perform better than ResNet with the same amount of computation, and it can lower the number of hyper-parameters while keeping accuracy high (Zhao et al., 2023).

The quality of intrusion detection classification depends on extracting data features and selecting parameters in the classifier. In traditional methods, these two aspects are optimized separately. However, with deep learning technology, both can be optimized simultaneously to improve classification performance. Although deep learning has been widely applied to intrusion detection in recent years, many existing methods still have several challenges. Firstly, most approaches rely on individual feature vectors as input, thereby neglecting potential correlations among different features in intrusion behaviors, particularly those between non-consecutive features. Secondly, while image-based processing methods offer novel perspectives for intrusion detection, their image generation processes are either simplistic or overly complex, leading to increased computational costs.

This paper proposes an intrusion detection method based on ResNeXt50, aiming to tackle the aforementioned challenges. It has three main parts: (1) processing the dataset; (2) changing the trunk convolution and using the multi-scale feature fusion module SPPM (Soft Pool (Stergiou et al., 2021) and Pyramid Pooling Module (Zhao et al., 2017)); and (3) adding focal loss (Lin et al., 2017) because of the uniqueness of the network flow data and the unevenness of the dataset to improve the model's final detection ability.

In summary, the main contributions of this paper are as follows:

- (1) A data-imaging process is introduced to convert network flow data into a two-dimensional image; this addresses the limitation of traditional methods, which fail to consider the spatial relationships among features, and it enables the model to identify potential relationships within the network flow data features.
- (2) Following the above data image processing, a classification model based on ResNeXt50 is suggested. This model uses aggregated residual blocks instead of basic residual blocks to calculate the feature maps, thereby enhancing its expressiveness without adding extra parameters. Furthermore, enhancing the model's performance by adjusting its backbone convolutional structure and integrating it with the multi-scale feature fusion module SPPM. Within this framework, soft pool effectively addresses the issue of feature map information loss encountered by the original

adaptive pooling layer, thereby enhancing the model's feature capture capability.

- (3) The model's loss function has been improved to focal loss to mitigate the impact of sample imbalances on classification.

The technology roadmap of this paper is shown in Fig. 1. The remainder of this paper is organized as follows: Section 2 describes the related research work, including data imageization processing methods and residual learning in the field of intrusion detection. Section 3 describes the theoretical background and specific improvements of the data-to-image method in this paper and introduces the core components of the proposed model. Section 4 conducts experimental analysis and evaluation and verifies this paper's method's reasonableness and effectiveness through multi-group comparison and ablation experiments. Section 5 summarizes this paper and outlines future research directions.

2. Related work

In recent years, deep learning has made remarkable achievements in image recognition, speech recognition, natural language processing, and beyond. Its excellent performance in processing complex data has led to innovative approaches for dealing with multi-feature intrusion detection data. Numerous researchers are actively introducing various neural networks into the realm of intrusion detection. These deep learning-based intruder detection algorithms have further propelled the advancement of the network intrusion field.

2.1. Some deep learning methods

Xu C et al. (Xu et al., 2018) proposed an intrusion detection technique that combines gate recurrent unit (GRU), multi-layer perceptron (MLP), and recurrent neural network (RNN) architectures with a softmax module. They experimentally confirmed the enhanced efficiency of the model in the field of intrusion detection. Liu H et al. (Liu et al., 2019) introduced an end-to-end detection method for analyzing payloads using deep learning models. They proposed classification methods based on convolutional neural networks (CNN) and recurrent neural networks (RNN) for attack detection. Lee J H et al. (Lee and Park, 2021) utilized a generative adversarial network (GAN) to address dataset imbalance problems by generating virtual data, coupled with a random forest (RF) for intrusion detection classification. According to Fu Y et al. (Fu et al., 2022), attention mechanisms are employed with bidirectional long short-term memory (Bi-LSTM) networks for intrusion detection, achieving high accuracy. However, the complexity of feature extraction

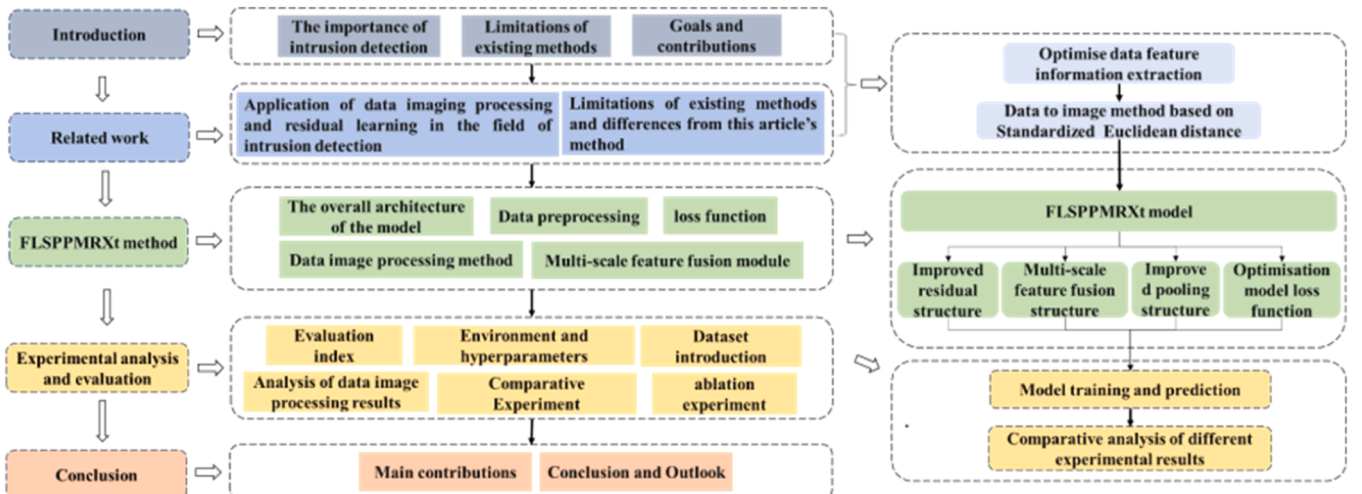


Fig. 1. This article's technology roadmap.

might reduce the model's ability to generalize. Cui J et al. (Cui et al., 2023) proposed the use of stacked autoencoder (SAE) modules for feature extraction, coupled with generative adversarial networks (GANs) integrating Gaussian mixture models for classification. Dina A S et al. (Dina et al., 2023) proposed the application of the focal loss function to convolutional neural networks to address sample imbalance issues in intrusion detection datasets. The above methods effectively apply deep learning to intrusion detection and propose methods to solve the data imbalance problem in network streams. However, most of the above approaches are vector learning of individual data stream features, which ignores the potential relevance of the intrusion data. Specifically, they are unable to capture the spatial relationship between non-adjacent features. In order to deal with this problem, this paper proposes a method of data-to-image processing.

In addition, Lu K D (Lu et al., 2025) et al. proposed a lightweight method, MODEO—CNN, based on multi-objective joint optimisation combining Multi-Objective Discrete Extreme Optimisation (MODEO), Neural Architecture Search (NAS), and Hyperparameter Optimisation (HPO). The method uses a hybrid variable-length coding strategy to characterise the network topology. It combines the principle of Pareto Optimality to achieve better detection accuracy on Gas Pipeline, BoT-IoT and Power System Attack datasets to achieve higher detection accuracy. Zeng G Q et al. (Zeng et al., 2025) proposed EvoAAE, an unsupervised anomaly detection method combining Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs). By tuning the hyperparameters and network architecture through improved Particle Swarm Optimization (PSO), they achieved higher accuracy in four IIoT datasets. The above methods use sequence encoding to improve model efficiency through automated search or adapt to unsupervised scenarios with scarce labels, but in this paper, the method break through the dependence of traditional sequence modelling methods on local features by transforming network traffic data into image representations that conform to the nearest-neighbour distributions of the features, and explicitly capture the spatial correlations of the non-adjacent dimensions in the attack features.

2.2. 2D CNN methods

To address the issue of dimension mismatch between intrusion detection data and deep learning models while maintaining the spatial correlation of features, numerous scholars have made advancements in the coding domain. These methods primarily involve mapping one-dimensional data onto two-dimensional images, followed by image classification processing.

Li Z et al. (Li et al., 2017) introduced a coding technique named grayscale coding, which converts digital features into 8-bit binary pixels for classification using an image classification model. In a subsequent study, Kim T et al. (Kim et al., 2018) expanded this method by applying it to pseudo-RGB images and employing convolutional neural networks for classification. However, while these methods offer innovative approaches to intrusion detection, they have limitations. Specifically, they can only map features into pixels in the order of their input, ignoring the spatial relationships among features. Consequently, the classification accuracy of the final model is not satisfactory (Liu et al., 2019).

According to Andresini G et al. (Andresini et al., 2021), an autoencoder (AE) was utilized to reduce the dimensionality, followed by k-nearest neighbor (KNN) and clustering algorithms to create an image representation and identify potential connections between adjacent data points. Finally, a CNN was employed to categorize the data into groups. In a separate paper (Andresini et al., 2021) authored by the same researchers, they used DeepInsight (Sharma et al., 2019), a novel approach for converting data into images. Subsequently, these images underwent processing with Auxiliary Classifier GAN (ACGAN) to generate samples for a few attack classes, thereby balancing the dataset and improving classification accuracy. These two methods improve the shortcomings of the above two parsimonious methods. They can effectively capture the

spatial correlation of features, but the image-processing steps are more complicated. The first method (Andresini et al., 2021) requires training the autoencoder and then performing the nearest neighbor search in order to complete the data image processing. The second method (Andresini et al., 2021) necessitates the coordinate mapping of the Cartesian coordinate system and employs the convex hull algorithm to determine the image size; additionally, it must address the intra-pixel collision issue (there is a case of multiple features within a pixel when generating an image). In this paper, the proposed method does not require complex operations. It can directly perform image transformation on the pre-processed data.

2.3. Residual learning methods

CNNs based on residual learning have achieved outstanding results in image recognition (Chen et al., 2019), and they are easier to train and optimize than ordinary CNNs. The superiority of residual learning in image classification and object recognition has been established in numerous studies (Feng et al., 2020; McAllister et al., 2018; Li et al., 2020). Therefore, many researchers have tried residual learning-based CNN to detect network intrusions.

Man J. et al. (Man and Sun, 2021) constructed a deep network model with residual blocks to achieve classification of network streaming data. Duan X. et al. (Duan et al., 2023) proposed a method to find anomalies that utilizes residual learning, wavelet transform, and multi-scale residual features to capture detailed information across different levels and enhance detection performance. Sun M. et al. (Sun et al., 2022) proposed a deep convolutional intrusion detection model that combines residual units and employs SMOTE to handle unbalanced datasets. Zhao G. et al. (Zhao et al., 2023) proposed an intrusion detection algorithm that leverages a generative adversarial network (GAN) to enhance minority samples. Subsequently, they employ a gated recurrent unit (GRU) to extract the time-series features of the samples. Finally, they utilize a residual network (ResNet) along with Softmax to categorize and normalize the results. The aforementioned research demonstrates the effectiveness of residual learning in the field of intrusion detection. This paper employs ResNeXt, which builds upon the foundation of residual learning. The model integrates group convolution with residual learning, enhancing its accuracy without altering the number of model parameters, thereby effectively improving its efficiency. The validity of the model is ultimately corroborated through experimental evaluation.

Table 1 summarizes the existing relevant research on network intrusion detection in this paper, including the technologies, data sets, and pros and cons used in the related work. This table provides a valuable reference for understanding the current research status of network intrusion detection technology.

2.4. Discussion

This paper introduces a novel approach for data-to-image conversion, aimed at enhancing the retention of spatial knowledge among features. It offers a compact representation of the image for model training and presents a method that combines multi-scale feature fusion modules with a deep learning model.

While focal loss, pyramid pooling module (PPM), and residual networks have been utilized in prior research, our approach stands out due to the specific combinations employed within these structures and their practical effectiveness when integrated. Moreover, the central focus of this paper is to validate the effectiveness and reliability of the proposed data-to-image transformation method. Indeed, the method proposed in this paper outperforms several other recent, related works in this field on the same dataset. The research in this paper demonstrates the feasibility of deriving a compact image representation of network flow data based on the similarity between features. This approach better preserves the spatial neighborhood structure of the features. Furthermore, this paper pioneers the use of soft pooling in the field of intrusion detection.

Table 1

Summary of existing methods for network intrusion detection: technology, data set, and pros and cons.

Paper	Technique	DataSet	Pros/Cons
Xu C et al. (Xu et al., 2018)	Use the GRU to store and extract features. MLP pairs were used for nonlinear classification decisions.	KDD 99 NSL-KDD	Automatic feature extraction. Multiple models were validated. High accuracy. Using old datasets. Using a 1D data stream that ignores relationships between disjoint features.
Liu H et al. (Liu et al., 2019)	Learning features from raw network packets. Extracting feature information using word embedding techniques. Proposing CNN and RNN-based classifiers.	CNTC-2017 Webshell DARPA1998 KDD99 CSIC-2010 HTTP	High accuracy in selected datasets. Model Complexity. Using a 1D data stream that ignores relationships between disjoint features.
Lee J H et al. (Lee and Park, 2021)	Generating Minority Class Images Using GAN. Using random forest classifiers.	CICIDS 2017	Solved the data set imbalance problem. High computational overhead.
Fu Y et al. (Fu et al., 2022),	Expanding datasets with adaptive synthetic sampling. Feature dimensionality reduction using a stacked autoencoder. A classifier was proposed based on attention mechanisms and bidirectional long-short-term memory (Bi-LSTM).	NSL-KDD	Using old datasets. Low accuracy. Using a 1D data stream that ignores relationships between disjoint features.
Cui J et al. (Cui et al., 2023)	Stacked autoencoders (SAE) modules are used to reduce the dimensionality and reconstruct the data. Clustering based on the Gaussian mixture model (GMM) was used to undersample most class samples. Enrich minority samples with GMM-based Wasserstein generative adversarial networks. Using CNN-LSTM-based classifiers.	NSL-KDD UNSW-NB15	Low accuracy. Complex models with high computational overhead. Using a 1D data stream that ignores relationships between disjoint features.
Dina A S et al. (Dina et al., 2023)	Using focal loss to cope with dataset imbalance. Using CNN and FNN as classifiers, respectively.	Bot-IoT WUSTL-IIoT-2021 WUSTL-EHMS-2020	An imbalance in the number of dataset categories was considered. Proof of the validity of the focal loss. Using a 1D data stream that ignores relationships between disjoint features.
Lu K D (Lu et al., 2025)	Using multi-objective optimisation methods based on Multi-Objective Discrete Extreme Optimisation (MOEEO), Neural Architecture Search (NAS), and Hyperparameter Optimisation (HPO).	Gas Pipeline BoT-IoT Power System Attack	Designed a lightweight convolutional neural network. High detection accuracy.

Table 1 (continued)

Paper	Technique	DataSet	Pros/Cons
Zeng G Q et al. (Zeng et al., 2025)	Using the new hybrid variable-length coding strategy. Designing particle swarm optimisation strategies with efficient binary coding strategies using unsupervised anomaly detection based on automatic adversarial deep learning.	SWaT, WADI MSL, PSA	Solving the problem of scarcity of tagged data in IIoT
Li Z et al. (Li et al., 2017)	Convert data to 8×8 grayscale images. Validation using ResNet and GoogLeNet as classifiers, respectively.	NSL-KDD	Using old datasets. Data-to-image methods are too simple and ignore the correlation between features. Low accuracy.
Kim T et al. (Kim et al., 2018)	Convert data to 12×12 RGB-like images. Using GooLeNet as a classifier.	KDD Cup 1999 UNSW-NB15 IDS2017	Correlation between features ignored in data-to-image conversion. Low accuracy.
Andresini G et al. (Andresini et al., 2021)	Representation of network streams as 2D images using nearest neighbor search and clustering algorithms. Feature dimensionality reduction using self-encoders. Using a custom CNN structural classifier.	KDDCUP99 UNSW-NB15 CICIDS2017	Feature correlation is considered in data-to-image conversion but is computationally complex. High accuracy.
Andresini G et al. (Andresini et al., 2021)	Representation of network streams as 2D images using the DeepInsight algorithm. Generating minority-class samples using ACGAN. Using a custom CNN structural classifier.	KDDCUP99 UNSW-NB15 CICIDS2017 AAGM17	Novel and effective data-to-image method, but computationally complex and time-consuming. A time-consuming model with high computational overhead. High accuracy.
Man J et al. (Man and Sun, 2021)	Convert the data to 14×14 images. Dealing with dataset imbalances using focal loss. Creating Deep Convolutional Neural Network Classifiers with Residual Blocks. Using batch normalization and global average pooling to avoid overfitting.	UNSW-NB15	Using resize shapes directly to convert data to images, ignoring the correlation between features. Low accuracy.
Duan X et al. (Duan et al., 2023)	Multi-scale generation of raw flows using sliding windows and wavelet transforms. Feature reconstruction using a stacked autoencoder. Constructing a multiplexed residual swarm network to capture feature information.	KDD99 NSL-KDD UNSW-NB15 CICIDS2018	Model Complexity. Low accuracy.
Sun M et al. (Sun)	Generating Minority Class Samples with Borderline-SMOTE.	NSL-KDD	Using old dataset.

(continued on next page)

Table 1 (continued)

Paper	Technique	DataSet	Pros/Cons
et al., (2022)	The data were converted to 32×32 grayscale maps using principal component analysis incorporating a bicubic interpolation algorithm. Creating CNN structural classifiers.		
Zhao G et al. (Zhao et al., 2023)	Balancing datasets using GAN. Extracting time-series features using GRU. Classification using residual networks.	NSL-KDD	Using old datasets. High computational overhead. High accuracy.

Experiments show that the combination of soft pool and multi-scale feature fusion algorithms significantly enhances intrusion detection classification accuracy.

3. FLSPPMRXt method

This section starts by explaining the structure of the ResNeXt-based model, which is proposed in this paper. It then discusses the data pre-processing and data visualization methods. Finally, it introduces the model's main part, which is the multi-scale feature fusion module with an improved pooling layer. The model structure of the intrusion detection method in this paper is shown in Fig. 2.

3.1. FLSPPMRXt architecture

In this paper's proposed FLSPPMRXt model, the initial convolution of the backbone network consists of one 5×5 convolution coupled with two 3×3 convolutions (the original backbone convolution in this model was a 7×7 convolution), which is determined based on the results of the ablation experiments in Section 4.6. After completing this step, the structure employs a multi-scale feature fusion module to enhance the model's feature extraction abilities. For this application, the model employs a grouped convolution with a channel count of four and a group count of 32. The model is activated using the sigmoid function after global average pooling and full connectivity. A description of the model

structure appears in Table 2.

The input of each sample into the model is achieved by processing the data into two-dimensional grayscale images. In the initial stage of convolution algorithms, features are extracted from grayscale images. After the output feature maps are fed into the module for fusing multi-scale features, fused feature maps containing different sensory fields are generated. Finally, the fusion feature maps are input into the grouped convolution module for training to yield the final classification results.

3.2. Data pre-processing

The characteristics of the intrusion detection dataset indicate that many data sets include character-based features such as protocols and services, as well as other label features. Since non-numerical data is not uniformly processed, it is necessary to process this non-numerical data. Table 3 provides information on the non-numerical data feature types and the appropriate processing operations.

The character features contained in the dataset must be numerically processed using one-hot coding, while the label features must be label-coded to obtain a dataset where all features are of a continuous type. Additionally, due to the varying scales and disparities among each column feature, it is necessary to normalize the data according to Eq. (1). In

Table 2
Structure of FLSPPMRXt model.

stage	output	ResNeXt-50($32 \times 4d$)
Conv1	9×9	5×5 , 64, stride 2, padding 4
Conv2	9×9	3×3 , 64, stride 1, padding 1
Conv3	9×9	3×3 , 64, stride 1, padding 1
Conv4	9×9	Multi-scale feature fusion module -SPPM
		$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128, C=32 \\ 1 \times 1, 256 \end{bmatrix} \times 3$
Conv6	5×5	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256, C=32 \\ 1 \times 1, 512 \end{bmatrix} \times 4$
Conv7	3×3	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512, C=32 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$
Conv8	2×2	$\begin{bmatrix} 1 \times 1, 1024 \\ 3 \times 3, 1024, C=32 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$
Conv9	1×1	Global average pool 2-d fc, sigmoid

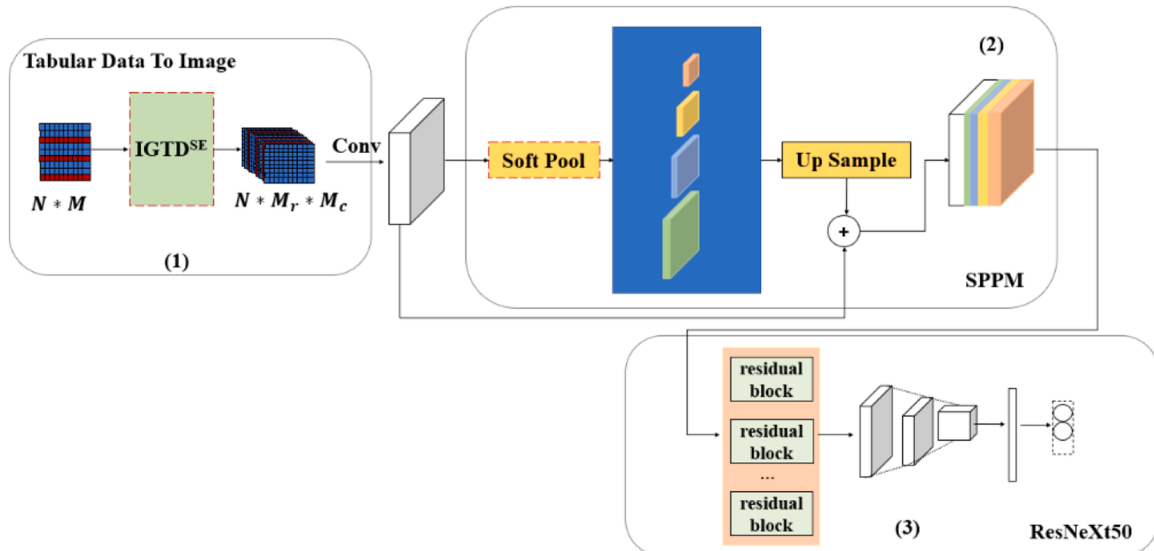


Fig. 2. Structure of FLSPPMRXt model.

Table 3

Types and operations of data features.

Feature Type	Operation
Character	One-Hot Encoding
Numerical	–
Categorical	Label Encoding

the equation, x represents each column feature, x_{min} denotes the minimum value of each column feature, and x_{max} refers to the maximum value of each feature in each column.

$$x^* = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

3.3. Improved data-to-image methods

This paper presents a data-to-image method by improving the algorithm of the Image Generator for Tabular Data (Zhu et al., 2021), comprising the following steps.

For the overall sample data X , let $X = N \times M$; where M is the amount of features and N is the number of samples, $X_{i,:}$, $X_{:,j}$, X_{ij} refer to the elements of row i , column j and row i , column j respectively. A calculation is made of the distance between features on a pairwise basis using a distance metric, in which small distances correspond to small ranks and large distances correspond to large ranks. This matrix F represents a matrix of feature distance rank. Assuming the target image is of size $M_r \times M_c$ (satisfying $M_r \times M_c = M$), treat the image as a grid. Assign coordinates to each pixel, compute the spatial distance between all pixel pairs, and sort these distances in ascending order to obtain a pixel distance rank matrix I . In both instances, F and I are $M \times M$ symmetric matrices, which can be observed in the final generation of rank matrices for feature distances and pixel distances in Fig. 3a and Fig. 3b. However, comparing the two figures, the two matrices cannot be directly matched because of their substantial differences, so it becomes necessary to define the error function as follows:

$$\text{err}(F, I) = \sum_{i=2}^M \sum_{j=1}^{i-1} \text{diff}(F_{ij}, I_{ij}) \quad (2)$$

With this function, we can measure and optimize two matrices, where F_{ij} indicates the rank value of the i th feature and the j th feature, I_{ij} contains the rank value of the i th pixel, and the j th pixel, and the $\text{diff}(:, :)$ function determines the distance between F_{ij} and I_{ij} either by absolute or squared.

Initially, the i -th feature is assigned to the i -th pixel. By swapping the pixel assignment order of features (i.e., reordering the feature sequence), the algorithm minimises $\text{err}(F, I)$ to make F as close as possible to I . It iteratively selects a feature that has not been swapped for the longest time, attempts to swap it with all other features, and chooses the

swap pair that yields the greatest error reduction. This process repeats until convergence, yielding an optimal feature ordering. The $M_r \times M_c$ image is then filled with feature values according to this ordering. In the final image, semantically similar features (i.e., those with small distances in F) are assigned to spatially adjacent pixels (i.e., those with small distances in I), forming a structured image with meaningful local patterns.

This method calculates the error of both the initial feature rank matrix and the pixel rank matrix. The feature rank matrix is optimized to reduce error by swapping the rank order of features iteratively. After error optimization, each sample X_i is finally transformed into an image of $M_r \times M_c$, where $M_r \times M_c = M$.

The distance in the above method is calculated using Euclidean distance, which treats differences between attributes (or measures) as equal. However, when dealing with intrusion detection data, there are large differences between different attributes, and using Euclidean distance may weaken some feature associations. Therefore, this paper improves it to a Standardized Euclidean distance, which is standardized according to Eq. (3) before performing the distance metric, where a value of m indicates the mean value and a value of s indicates the standard deviation.

$$X^* = \frac{X - m}{s} \quad (3)$$

This paper proposes a data visualization method that employs a standardized Euclidean distance to determine the initial ranking matrix for feature distances and the ranking matrix for pixel distances. The larger the distance, the higher the rank, and the darker the color of the point, the more significant it becomes. As demonstrated in Fig. 3c, the feature rank matrix has been optimized by taking into account the pixel rank matrix, resulting in a closer alignment between the optimized feature distance rank matrix and the matrix of pixel distance ranks compared to the original matrix.

This article's data image processing methods utilize three distance measures: Euclidean distance, cosine distance, and Standardized Euclidean (SeEuclidean) distance. All three distance measures can be used to measure the similarity between features. Their formulas are shown in Eqs. (4), (5), and (6), respectively, where x_i and y_i denote the coordinates of different points in a particular dimension, respectively.

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (4)$$

$$\cos(x, y) = 1 - \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}} \quad (5)$$

$$d(x, y) = \sqrt{\sum_{i=1}^n \left(\frac{x_i - y_i}{s_i} \right)^2} \quad (6)$$

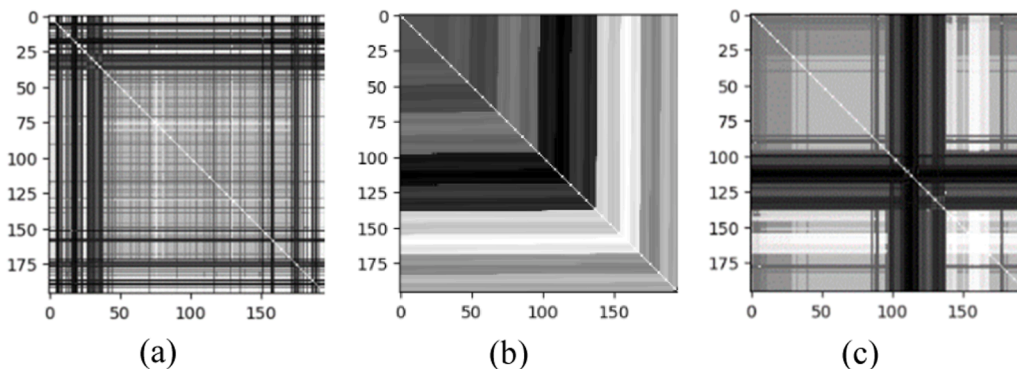


Fig. 3. Correlation matrix of normalized Euclidean distance generated images: (a) Feature rank matrix, (b) Pixel rank matrix, (c) Optimized feature matrix.

Fig. 4 illustrates the images generated by each technique on UNSW_NB15 dataset; Fig. 4a and Fig. 4b depict the image created by Euclidean distance; Fig. 4c and Fig. 4d exhibit those derived from cosine distance; and Fig. 4e and Fig. 4f demonstrate the graphics generation by Standardized Euclidean distance. In the process of image generation, the distance metric serves as a measure of similarity between two pairs of features (represented by distance). The optimization process aims to increase similarity (i.e., decreasing distance) and ultimately generate image representations with tightly arranged similar features. According to the figures, the image generated by Standardized Euclidean distance exhibits a more compact feature representation (feature neighbor representation with high similarity).

The data-to-image method proposed in this paper optimizes the image transformation by minimizing the difference between the feature distance ordering of tabular data and the pixel distance ordering of the generated image. Each feature corresponds to one pixel, resulting in similar features being displayed as closely adjacent pixels in the image. This approach thereby facilitates the generation of images from network streaming data.

3.4. Improved multi-scale feature fusion module

For deep learning networks, as the network becomes deeper, the receptive field also increases, which can hinder the network's ability to effectively fuse global feature information, and PPM solves this problem. However, the global average pooling (GAP) employed in PPM tends to lose information during the compression process and has limited capabilities for fusing and extracting information. Therefore, this paper proposes an improvement by upgrading the pooling layer to a soft pool.

Soft pooling is a technique that retains the basic input attributes by utilizing a SoftMax-based weighting system during the upscaling of more intense feature activation values, as demonstrated in the following equation:

$$\tilde{\mathbf{a}} = \sum_{i \in \mathbf{R}} \frac{e^{a_i} * \mathbf{a}_i}{\sum_{j \in \mathbf{R}} e^{a_j}} \quad (7)$$

Where $\mathbf{a}$ represents the activation mapping of the region, $\frac{e^{a_i}}{\sum_{j \in \mathbf{R}} e^{a_j}}$ represents the activation weight, which is the sum of the natural exponential of the activation value divided by the natural exponential of all activation values, and the final output $\tilde{\mathbf{a}}$ is the weighted sum of all weights and the activation mapping $\mathbf{a}$. The soft pooling accumulates activations in an exponentially weighted manner, and all activations within the convolution kernel contribute to the final result. The importance of activations varies depending on the weight, with higher activations indicating more significant characteristics of the feature map. By using this method, the down-sampled activation mapping can improve the accuracy of the final classification by retaining more information.

Fig. 5 provides a more detailed illustration of the improved multi-scale feature fusion module. The enhanced SPPM module is capable of effectively aggregating contextual information from diverse regions, thus facilitating the model's acquisition of global information.

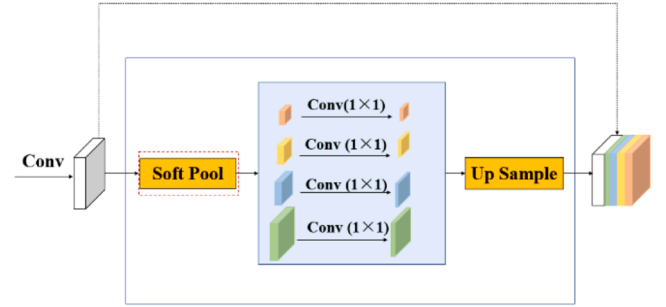


Fig. 5. Structure of the SPPM module.

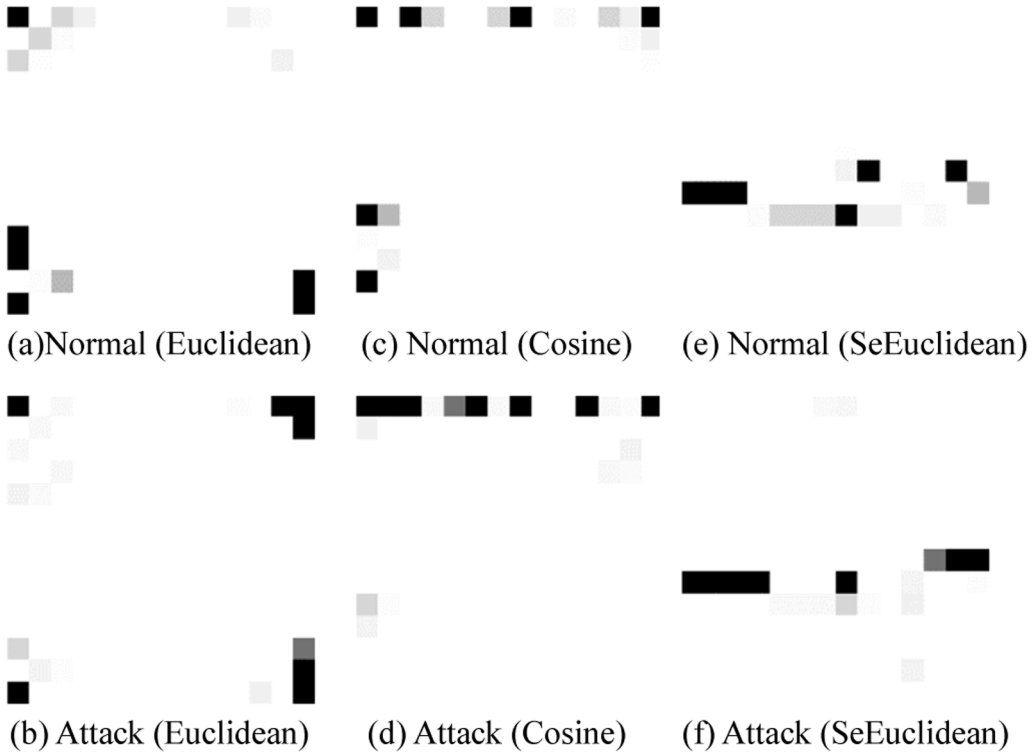


Fig. 4. Generate attack and normal images in different ways: (a) Normal image generated by Euclidean distance, (b) Attack image generated by Euclidean distance, (c) Normal image generated by Cosine distance, (d) Attack image generated by Cosine distance, (e) Normal image generated by SeEuclidean distance, (f) Attack image generated by SeEuclidean distance.

First, the model extracts features through the combined convolution of the backbone. Next, the SPPM module extracts features at different scales. Finally, the soft pooling generates four feature maps with different sizes (1, 3, 5, and 7). Subsequently, each of these feature maps is down-sampled using 1×1 convolution to ensure the correct number of channels. Next, each feature map is up-sampled to its original size, and finally, the channel dimensions are concatenated to obtain the multi-scale feature-fused feature maps. Compared to the original Max Pooling algorithm, the SPPM module incorporates feature information from diverse sensory fields and sub-regions, while the application of soft pooling further enhances the features and boots the model's performance.

3.5. Loss function

By adjusting the internal weights, focal loss prevents a few samples from poorly fitting and negatively impacting classification results. The method involves adding a weight adjustment factor of $(1-p)^\gamma$ to the cross-entropy loss function to offset the effects of imbalances in the number of rare sample categories or classification difficulty. The formula can be stated as follows:

$$L_i = \begin{cases} -\alpha(1-p)^\gamma \log(p) & y = 1 \\ -(1-\alpha)p^\gamma \log(1-p) & y = 0 \end{cases} \quad (8)$$

The hyperparameter γ serves as a positive focus parameter, adjusting the weights of both easy and hard-to-classify samples to direct the model's attention towards the latter. The hyperparameter α functions as a balancing parameter, compensating for the disparity in the influence of positive and negative samples on the loss values. Focused loss lowers the loss contribution of easy-to-score samples by changing the weight share between difficult and easy samples in the loss function. This approach is beneficial because it simplifies the model's task of detecting abnormal traffic patterns, eliminating the need for complex weight mappings.

4. Experimental analysis and evaluation

The general flow chart for our model's detection of abnormal network data traffic is illustrated in Fig. 6.

- (1) Data pre-processing: numerical processing of the character features of the original data, execution of one-hot encoding and label encoding operations, followed by data normalization.
- (2) Data-to-Image: The processed data are mapped from one-dimensional to two-dimensional image representation; a gray-scale image is constructed for each network data, and its corresponding labels are generated.

- (3) Classification model: A classification model FLSPPMRXt based on ResNeXt50 is constructed, and the training set is redivided into a training set and a validation set to train and tune the model for intrusion detection purposes.

4.1. Evaluation metrics

To evaluate model performance in this paper using overall classification accuracy (OA), recall (R), and F1 score (F1). OA shows intrusion detection systems's capability of recognizing samples accurately. Recall refers to the capability of an intrusion detection system to detect all attacks. F1 represents the harmonic mean of precision(P) and recall(R). A higher F1 score generally indicates a better balance between accuracy and completeness.

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

$$R = \frac{TP}{TP + FN} \quad (10)$$

$$P = \frac{TP}{TP + FP} \quad (11)$$

$$F1 - Score = 2 \cdot \frac{P \cdot R}{P + R} \quad (12)$$

Where TN represents the number of samples that are normal and correctly identified; TP represents the number of samples that are attacked and correctly identified; FN represents the number of samples that are attacked but identified as normal; and FP represents the number of samples that are normal but identified as attacked.

4.2. Environment and hyperparameter settings

The proposed model's effectiveness has been validated on the NVidia GeForce RTX 3050Ti GPU model, operating under Windows. TensFlow1.14 and PyTorch1.10 were used to build the model, and the loss function's hyperparameters were set to $\gamma = 2$, $\alpha = 0.25$. In the training process, a batch size of 64 was used, and optimization was conducted using the Adam optimizer with a learning rate of $1e-4$.

To systematically select the optimal hyperparameters and evaluate their impact on model performance, this paper conducts tuning experiments on two key hyperparameters during the training process: batch size and learning rate. The experimental results are shown in Table 4. As seen from the table, there are some differences in model performance under different combinations of batch size and learning rate. Among them, when the batch size is 64 and the learning rate is $1e-4$, the model achieves the highest overall accuracy (93.2 %) and F1 score (95.2 %),

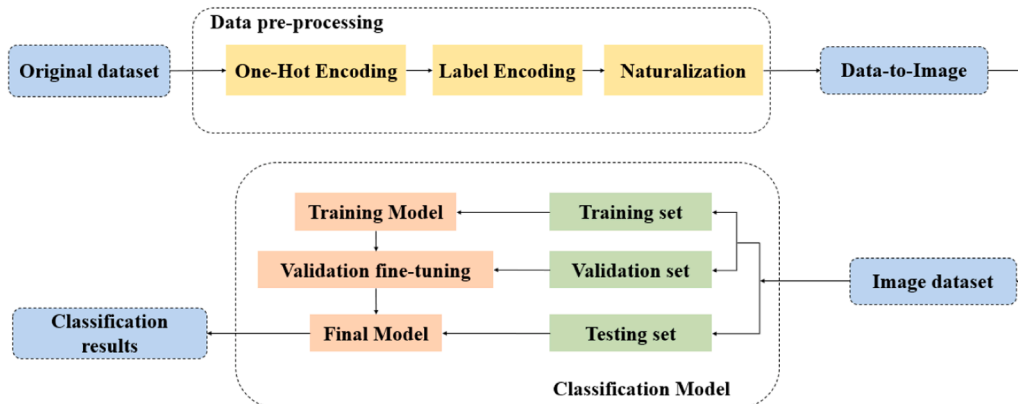


Fig. 6. Overall flow chart of FLSPPMRXt.

Table 4

Experimental results for different hyperparameter configurations.

batchsize	lr	OA	F1	R
128	1e-4	92.2	94.1	92.1
	1e-3	92.7	94.8	98.1
64	1e-4	93.3	95.2	96.3
	1e-3	92.6	94.8	99.5
32	1e-4	92.3	94.3	94.1
	1e-3	91.8	93.8	91.1

indicating that the combination has strong generalisation ability while maintaining good convergence. In addition, the recall rate of 96.3 % for this configuration suggests that the model also performs well in identifying attack samples.

In contrast, a larger learning rate (1e-3) improves the F1 score in some settings. Still, the overall stability is slightly worse, especially at smaller batch sizes (e.g., 32), where a significant performance degradation occurs. Larger batch sizes (e.g., 128) are more efficient, but fail to achieve better classification performance than smaller batch sizes in most cases. Therefore, based on its comprehensive advantages in multiple evaluation indexes, batch size = 64 and learning rate = 1e-4 are chosen as the default hyperparameter configuration of the model in this paper.

4.3. Dataset introduction

This paper validated the model's effectiveness using the publicly available dataset. The UNSW-NB15 (Moustafa and Slay, 2015; Moustafa and Slay, 2016) dataset consists of 44 feature attributes, including three character feature attributes, one label attribute, and 40 numerical attributes. These groups of features cover a wide range of attack types and include real data on everyday activity on modern networks. They better reflect the properties of modern network traffic data, making them suitable for testing the performance of intrusion detection models. As shown in Table 5, the training and test sets in the UNSW-NB15 dataset contain 82,332 and 175,341 samples, respectively. The feature representations of the dataset are presented along with the percentage and number of normal and attack samples. After data-to-image processing, the pre-processed data will be converted into 14×14 images. The training set will be divided into training and validation sets according to 7:3. A total of 57,632 sample sets will be used for training, while 24,700 samples will be in the validation set.

The NSL KDD dataset has 42 features, including three character-based features, 36 numerical features and one labeled column. The pre-processed data will be processed into 11×11 images, the training set and validation set will be split according to the aforementioned rules, and then the classification process will be carried out.

4.4. Analysis of data-to-image results

The Image Generator for Tabular Data algorithm assigns each feature to a pixel in the image. It achieves this by minimizing the pairwise

Table 5

Description and feature type of the UNSW_NB15 and NSL_KDD data.

		UNSW_NB15	NSL_KDD
Feature Type	Total	44	42
	Character	3	3
	Numerical	40	36
	Categories	1	1
Data Distribution	Train set	Total	82,332
		Normal	37,000 (44.9 %)
		Attack	45,332 (55.1 %)
	Test set	Total	175,341
		Normal	56,000 (31.9 %)
		Attack	119,341 (68.1 %)

distance between characteristics and pixels. This method spreads similar features to adjacent pixels to maximize feature correlation. Meanwhile, DeepInsight, another data-to-image algorithm, generates images using T-SNE projection and cannot represent features in the image using pixels. The tabular image generator produces more feature-compact images compared to the DeepInsight method and requires less training time. Fig. 7d compares the two methods in terms of training time per round.

By utilizing the improved Image Generator for Tabular Data, it will be possible to overcome ignoring feature differences in the distance metric algorithm. To determine the optimal method for generating the image dataset required for this study, we will compare the performance of three different metrics in image generation: Euclidean, Cosine, and Standardized Euclidean. Figs. 7a, 7b and 7c illustrate the variations in error that occur when using the three methods on different datasets for image generation.

Fig. 7a, 7b, and 7c illustrate how the error value changes when generating images for Euclidean distance, Cosine distance, and Standardized Euclidean distance on different datasets, respectively. The subsequent graph demonstrates that matrix error values decrease as iterations increase. Fig. 7e and 7f represent the final error and the comparison of the time taken to generate the images using the three approaches for different datasets, respectively. From Fig. 7, it is evident that the Standardized Euclidean distance exhibits the most minor error value and requires the least amount of time on the same dataset.

To evaluate the effectiveness of the proposed data-to-image conversion method, comparative experiments were conducted using a simple 2D-CNN model. The experimental results of the model on images generated by the proposed method and the DeepInsight method are summarized in Table 6. On the NSL-KDD dataset, the proposed method improved 3.73 % overall accuracy (OA) and 3.84 % in F1-score compared to DeepInsight. A more pronounced performance gain was observed on the UNSW-NB15 dataset, where the OA increased by 20.91 %, and the recall reached 85.6 %.

This paper further validates the superiority of standardized Euclidean distances by training and testing the three approaches on different datasets using the simple 2DCNN model proposed in (Andresini et al., 2021). Each result is displayed in Fig. 8. Fig. 8a and 8b show the test results on the UNSW_NB15 and NSL_KDD datasets, respectively. The findings reveal that the OA, recall, and F1 scores for images generated using Standardized Euclidean distance exceed those achieved by the other two methods.

To compare the effectiveness of various generation methods, this paper compares the error rates and time consumption of three distance metric methods by utilizing a simple CNN model structure. Following the above analysis, it is evident that the use of Standardized Euclidean distance for image generation is both accurate and time-efficient. Therefore, further experiments will be conducted utilizing images generated by this method.

As shown in Fig. 9, the experiment also compared the overall classification accuracy, recall, and F1 score of images on UNSW_NB15 generated using the Standardized Euclidean distance in both simple 2DCNN models and more complex models such as ResNet50 and ResNeXt50. The ResNeXt50 model outperformed the other two models in all three indices, indicating its suitability as a benchmark model.

4.5. Comparison test

The comparison test aims to assess the performance of the FLSPPMRxt model by comparing it with other related works in the field, utilizing the same dataset. This approach ensures that the evaluation is unbiased and minimizes the influence of different data samples on the results. For the UNSW_NB15 dataset, among the methods considered for comparison are one-dimensional convolution (Lopez-Martin et al., 2019), two-dimensional image convolution (Andresini et al., 2021; Man and Sun, 2021; Zhang et al., 2020; Al-Turaiki and Altwaijry, 2021), long

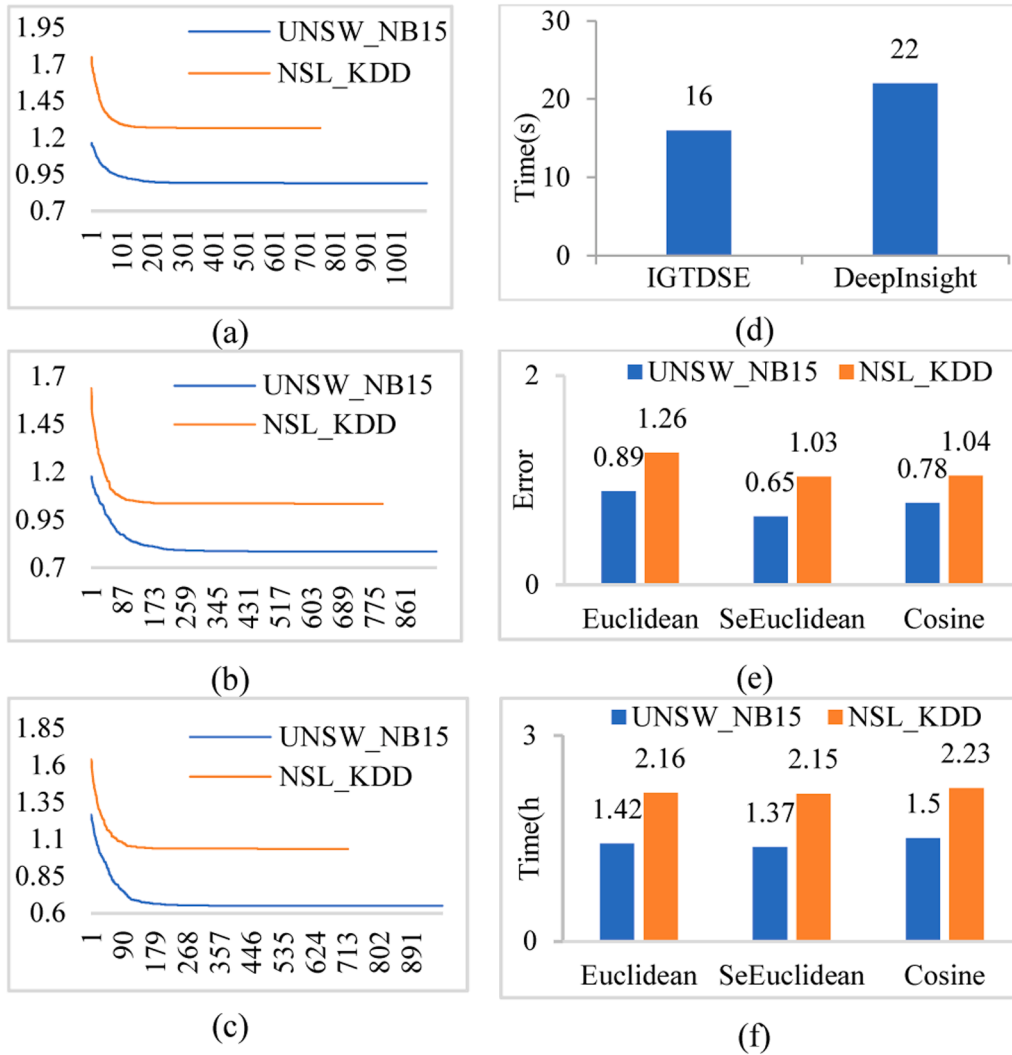


Fig. 7. Training time and error variation of different methods on different datasets: (a) Euclidean error, (b) Cosine error, (c) SeEuclidean error, (d) Training time in two different ways, (e) error variation, (f) Time to generate images.

Table 6

Comparison of results of different data to image methods on different datasets.

DateSet	Method	OA	F1	R
NSLKDD	DeepInsight	77.47	77.56	–
	Our Method	81.2	81.4	72.3
UNSW_NB15	DeepInsight	68.29	81.10	–
	Our Method	89.2	91.5	85.6

and short-term memory networks for network intrusion detection (Halbouni et al., 2022), feedforward neural networks (FNN) or deep neural networks (DNN) (Lin et al., 2017; Yang et al., 2020; Halbouni et al., 2022; Haitao et al., 2023), and machine learning techniques for network intrusion detection (ZHANG et al., 2022). Table 7 shows the results of testing this paper's method on the UNSW_NB15 and NSL_KDD datasets and the results of each metric for related studies under the same task.

In related comparative work, overall classification accuracy, F1 score, recall, and precision are used as evaluation metrics and cover many models such as traditional machine learning, deep learning, and hybrid architectures. The FLSPPMRXt model proposed in this paper performs well in all evaluation metrics. For the UNSW-NB5 dataset, its Overall Accuracy (OA) reaches 93.3 %, significantly higher than other methods. The F1 score of 95.2 % is at the top of the list, which indicates

that it performs well in balancing the recall and precision, and the recall and precision reach 96.3 % and 94.0 %, which are at the top of all the models, respectively. The results indicate the model performs well with low leakage detection and false alarm rates. For the NSL_KDD dataset, based on the OA metric, FLSPPMRX slightly outperforms SYN-GAN and is the best performer among all models. In addition, although SOM-KINN has the highest OA, its F1 score is significantly low, indicating that there is a particular category recognition bias; while traditional methods such as MLP and Naïve Bayes are relatively underperforming, reflecting the advantages of deep learning models in processing complex and high-dimensional data. Overall, this paper's approach is more competitive in intrusion detection tasks, proving the feasibility of this approach.

In addition, specific traditional machine learning methods (e.g., RF, MLP, etc.) perform well on specific metrics, but the overall performance is much lower than that of deep learning models. However, machine learning and deep learning algorithms ignore the spatial correlation between network flow features in these comparative studies. In contrast, the improved data image-based processing method considers the correlation between features as much as possible by reducing the distance error between features for imaging processing, which is the advantage of this method, and the final experimental results also prove the reasonableness of the proposed theory.

The experimental results presented in this paper confirm the

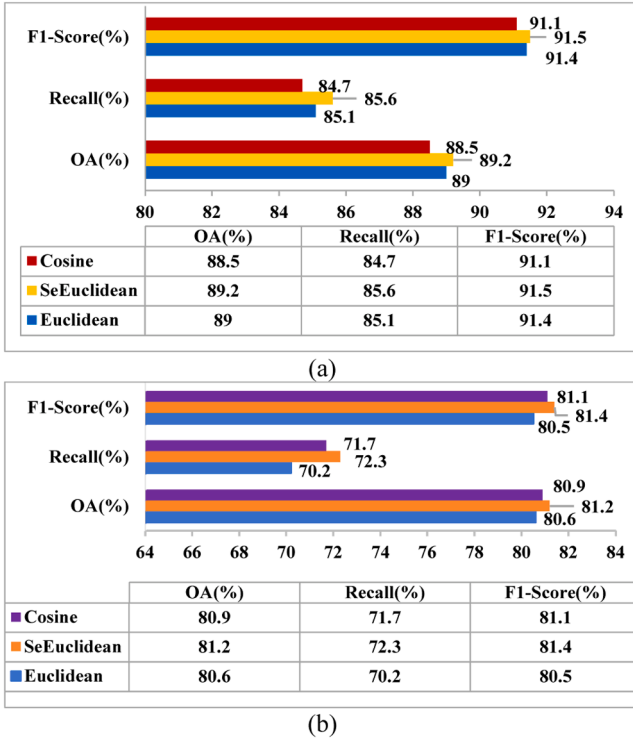


Fig. 8. 2DCNN classification OA, Recall, and F1 score for different methods: (a) UNSW_NB15, (b) NSL_KDD.

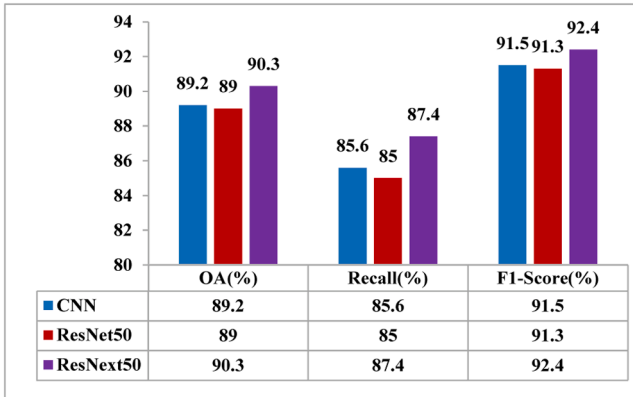


Fig. 9. Classification OA, Recall, and F1 score of three models.

reliability and validity of the proposed approach, which involves initially mapping network stream data into two-dimensional images and subsequently utilizing these images as model inputs for training the classification model. Although this concept has been explored in previous studies, this paper introduces a novel method of mapping data into images. The proposed data imaging processing method addresses the limitations of the previous methods (Li et al., 2017; Zeng et al., 2025), which overlooked the spatial correlation of features (Liu et al., 2019) also refers to it as a weakness). Specifically, the imaging processing takes into account the potential relationship between non-adjacent features, offering simplicity in operation. It enables image processing of network flow data without the need for complex model training, facilitating integration with existing model structures, and enhancing the efficiency of intrusion detection. Furthermore, this research introduces a model structure that exhibits superior feature extraction capabilities. As shown in Table 7, the proposed method exhibited superior accuracy in several sets of comparative experiments, outperforming other 2D CNN methods

that also focus on processing data streams into images. The data transformation to image method proposed herein, as well as the model's performance on different datasets, is sufficiently superior.

4.6. Ablation experiments

The ablation experiments first confirmed the effectiveness of the enhanced combined convolution. Subsequently, this study employed various padding values to assess the impact of varying initial feature map sizes on the final outcomes. Fig. 10 illustrates the fluctuations in loss and accuracy curves across different approaches on the UNSW_NB15 validation dataset.

In Fig. 10, it can be seen that there is less oscillation and an efficient convergence rate when combining convolution with a feature map size of 9×9 . The experimental validation results for the test set are presented in Table 8, which compares the results obtained using different methods of test validation. Among the first rows, the convolution kernel is 7×7 , and the corresponding feature map size is 7×7 . The remaining rows show the performance metrics of the convolution structure with different output sizes, with a convolution kernel of 5×5 and different combinations of stride and padding, and output feature maps with sizes of 7×7 , 8×8 , and 9×9 , respectively.

Based on the results of the experiments, subsequent experiments will use the ResNeXt50 model structure as a standard to extract initial features by combining convolution with an output size of 9×9 . Several ablation tests have demonstrated the effectiveness of the proposed multi-scale feature fusion module, SPPM, and the focal loss method. The experimental results on the UNSW_NB15 and NSL_KDD datasets, respectively, are presented in Fig. 11.

Fig. 11 depicts a set of original experiments followed by five sets of ablation experiments, arranged from bottom to top. The first set aims to assess the effectiveness of focal loss, while the test set is devoted to evaluating the ultimate model's performance. A comparison of the first set of experiments with the original experiments reveals that focal loss can enhance model accuracy to a certain extent. Additionally, Fig. 12 illustrates the fluctuation patterns of both loss and accuracy on the UNSW_NB15 validation dataset. This graph demonstrates that employing focal loss attenuates the oscillation of accuracy and loss values within the validation set.

The second and third sets of experiments verify that the binary cross-entropy (BCE) loss module works effectively when combined with the PPM and SPPM modules. As shown in Fig. 11, the overall accuracy and F1 score of the test set in the second set of experiments improved after using the PPM module compared with the original experiments. After using the improved SPPM module in the third set of experiments, the model's overall classification accuracy, recall rate, and F1 score all improved compared to the first set of experiments. This shows that the ideas in this paper work.

Moreover, a comparison between the second and third groups of experiments reveals that the effect is significantly better when the SPPM module is used. This demonstrates that SPPM, a multi-scale feature fusion module enhanced by integrating soft pooling, is superior in extracting features compared to the original PPM module. Soft pooling enables the preservation of more feature details during multi-scale feature extraction, thereby providing more feature information for the final classification.

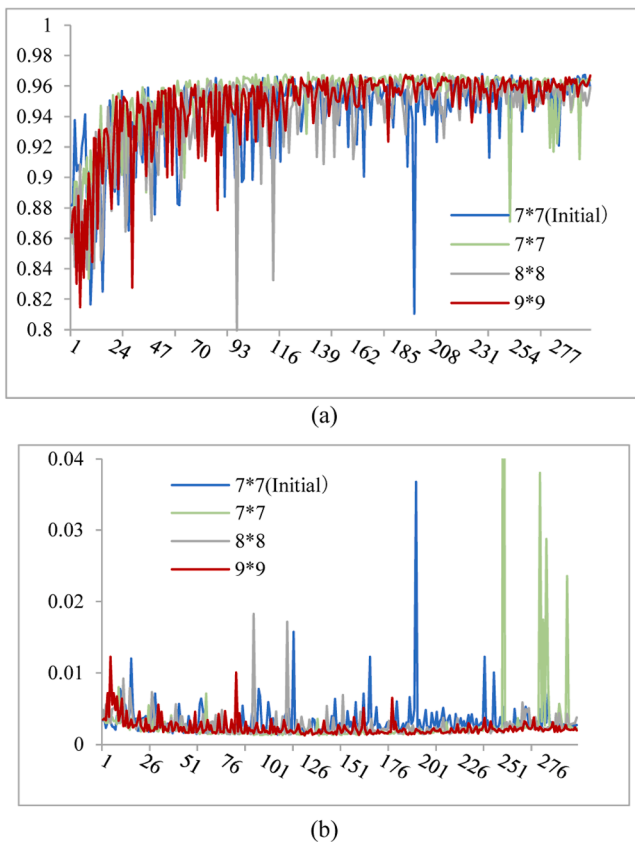
In the fourth and fifth experimental groups, it was confirmed that the combinations of FL with PPM or *sppm* were effective. Fig. 11 demonstrates that both FL and PPM, as well as FL and SPPM, improved model performance compared to the original model. Nevertheless, the combination of the SPPM module and focal loss exhibited the highest overall classification accuracy and F1 scores, with a recall rate on the UNSW_NB15 dataset reaching a high level of 96.3 %, leading to a superior classification result.

Overall, the SPPM structure performs optimally in most metrics, especially when combined with BCELoss and FocalLoss to achieve high

Table 7

Results of this paper on the UNSW_NB15 and NSLKDD datasets and related work.

Dataset	Methods	Type	Year	OA (%)	F1 (%)	R	P
UNSW_NB15	FLSPPMRXt	2D-CNN	2023	93.3	95.2	96.3	94.0
	TSDL (Khan et al., 2019)	AE	2019	89.7	87.0	92.5	86.7
	CNN-1D (Lopez-Martin et al., 2019)	1D-CNN	2019	89.8	91.3	98.0	85.5
	SAVAER-DNN (Yang et al., 2020)	FNN+DNN	2020	93.0	93.5	91.9	–
	MSCNN (Zhang et al., 2020)	2D-CNN	2020	91.4	–	–	–
	Pelican (Wu et al., 2020)	DNN (residual)	2020	86.64	–	–	–
	RLF-CNN (McAllister et al., 2018)	2D-CNN (residual)	2021	88.7	–	88.1	–
	MAGNETO-GAN (Andresini et al., 2021)	2D-CNN+GAN	2021	89.7	92.0	–	–
	BCNN (Al-Turaiki and Altwaijry, 2021)	2D-CNN	2021	90.3	90.5	90.0	91.0
	BCNN-DFS (Al-Turaiki and Altwaijry, 2021)	2D-CNN	2021	89.3	89.00	89.0	89.0
	SAE-ELM-Ca (Kushwah and Ranga, 2022)	SAE-ELM	2022	89.2	89.2	–	–
	ICVAE-DNN (Halbouni et al., 2022)	DNN	2022	89.1	–	–	–
	ESWRF (ZHANG et al., 2022)	RF+SMOTE	2022	78.0	76.0	78.0	80.0
	MSRC (Li et al., 2020)	SAE+ multi-channel residual block groups	2023	–	91.5	90.9	92.1
	SEMI-GRU (Haitao et al., 2023)	FNN+GRU	2023	88.1	88.1	81.6	96.3
	XGBoost-Simple-RNN (Kasongo, 2023)	RNN	2023	87.1	90.0	–	–
	LSTM (Kasongo, 2023)	LSTM	2023	80.1	90.4	–	–
	GMM-WGAN Module (Cui et al., 2023)	CNN+LSTM	2023	87.7	85.4	86.6	88.6
	IGRF-RFE (Yin et al., 2023)	MLP	2023	84.2	82.9	–	–
NSL_KDD	FLSPPMRX	2D-CNN	2023	84.3	85.4	80.3	91.1
	CRIPM-DM (Yuliana et al., 2024)	Cross-Industry Standard Process for Data Mining	2024	80	81	97	69
	SOM-KINN (Iftikhar et al., 2025)	KNN	2024	84.4	63.2	63.0	–
	MLP (Iftikhar et al., 2025)	MLP	2024	72.7	71.9	83.4	–
	SYN-GAN (Rahman et al., 2024)	GAN	2024	84.0	84.0	84.0	85.0
	Naïve Bayes (Kumar and Nagalakshmi, 2024)	–	2024	74.4	–	–	–

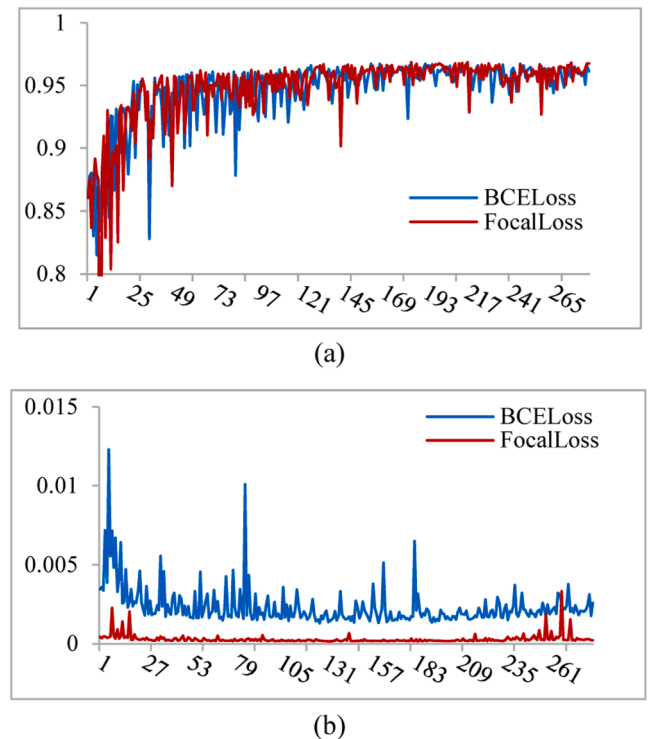
**Fig. 10.** Accuracy and loss values of models on UNSW_NB15 dataset trained in different ways versus size: (a) Different methods and sizes of accuracy, (b) Different sizes and methods of loss.

classification performance. Specifically, for the UNSW_NB15 dataset, SPPM+FocalLoss achieves an accuracy of 93.3 %, with a high recall of 96.3 % and an optimal F1 score of 95.2 %. For the NSL_KDD dataset, SPPM+BCELoss achieves an overall accuracy (OA) of 84.7 % with an F1

Table 8

Different initial convolution methods' OA, Recall, and F1 score.

Feature map size	kernel	Stride	Padding	OA	R	F1
7×7 (Initial)	7×7	2	3	90.3	87.4	92.4
7×7	5×5	2	2	90.4	89.3	92.7
8×8	5×5	2	3	89.7	87.4	92.0
9×9	5×5	2	4	90.6	88.5	92.8

**Fig. 11.** OA, Recall, and F1 score for different modality combinations on different datasets.

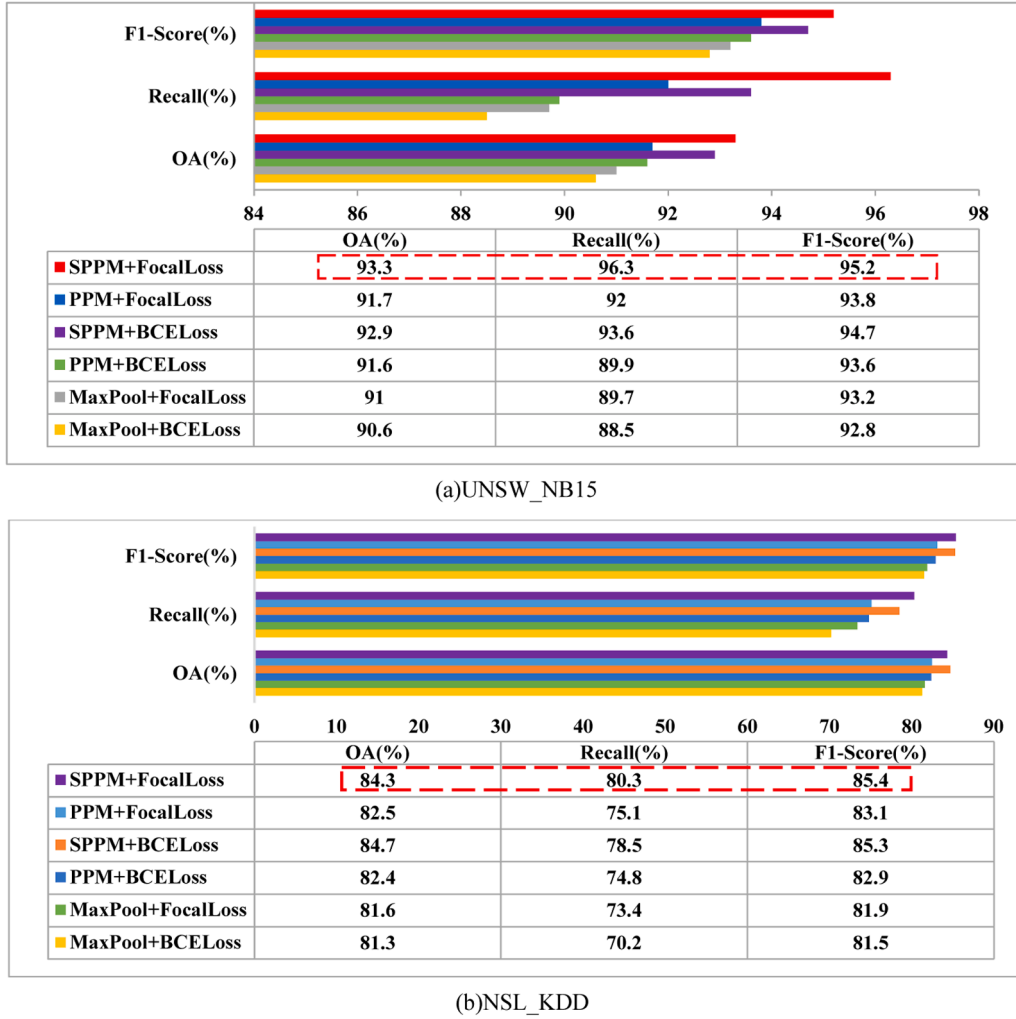


Fig. 12. Accuracy and loss on UNSW_NB15 dataset values of different loss functions: (a) Accuracy Comparison, (b) Loss value comparison.

score of 85.3 %. SPPM+FocalLoss improves its recall to 80.3 % with an F1 score of 85.4 %, showing its stronger ability to identify the attack class samples.

When comparing different ablation experiments, the proposed model structure outperforms the original classification model in terms of classification accuracy. This demonstrates the feasibility and effectiveness of combining all the components proposed in this paper.

5. Conclusion

Methods based on deep learning can uncover intricate intrinsic feature connections at an abstract level, greatly enhancing detection efficiency, reducing false positives and missed detections, and improving the overall accuracy of detection.

This paper proposes an intrusion detection method, FLSPPMRxt, which is based on ResNeXt50. This method uses a better data imaging processing algorithm that can turn one-dimensional network flow data features into a feature-compact image representation. This provides deeper learning models with a better input format and enables the model to learn how network flow data might be spatially related. Simultaneously, the proposed method fine-tunes the backbone convolution structure of ResNeXt50. Then, it integrates a multi-scale feature fusion module that enhances the pooling layer, thereby significantly improving the model's classification accuracy. The utilization of soft pooling enhances the model's ability to capture feature information. The model employs focal loss to alleviate the bias towards normal samples in the

intrusion detection classification model, thereby addressing the challenge of detecting attack samples. This mitigates the impact of sample category imbalances on categorization. Given the complexity and imbalance of network flow data in real-world networks, the proposed method effectively captures richer information pertaining to the characteristics of the data flow. Consequently, the model possesses additional information for data flow classification and addresses the issue of sample imbalance during training. This approach facilitates the development of a more generalized model capable of accurately predicting unseen.

Experiments conducted using UNSW_NB15, a widely recognized dataset in the field of intrusion detection, validate the effectiveness of the proposed model. Finally, numerous ablation experiments and comparisons demonstrate the paper's method over other techniques achieves better classification performance with accuracy and F1 scores as high as 93.3 % and 95.2 %, respectively.

Although the model has effectively distinguished between the categories of network flow data, several challenges remain to be addressed. While the UNSW_NB15 dataset is indeed complex, it still falls short of replicating real-world network traffic, particularly as intrusion detection aims to achieve real-time monitoring, which imposes stringent requirements on the speed and efficiency of the model. Therefore, in future endeavors, we aim to identify datasets with broader actual traffic coverage, thereby enhancing the model's generalization capabilities and effectiveness. In addition, there's a need to optimize the model in a more lightweight direction, striking a balance between performance and

efficiency in practical deployments.

CRedit authorship contribution statement

Jinxian Zhao: Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Haidong Hou:** Writing – original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Liang Chang:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Jinxian Zhao reports financial support was provided by Heilongjiang Province Department of Education.

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Data availability

The data that support the findings of this study are openly available in UNSW Canberra at ADFA at <https://cloudstor.aarnet.edu.au/plus/index.php/s/2DhnLGDdEECo4ys>.

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